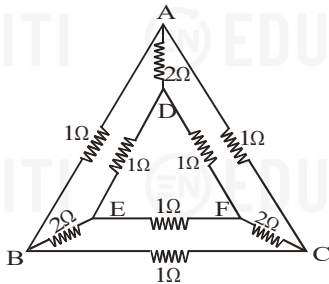
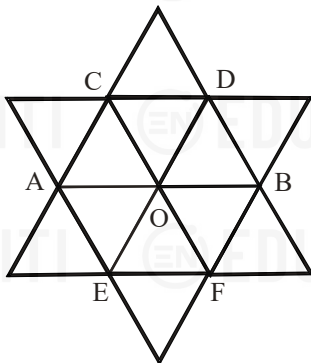


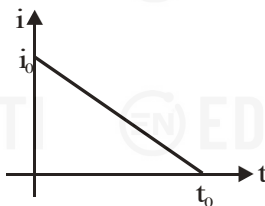
1. A network of nine conductors connects six points A, B, C, D, E and F as shown in figure. The figure denotes resistances in ohms. Find the equivalent resistance between A and D.



2. Find the equivalent resistance of the circuit between points A and B shown in figure. The resistance of each branch is R

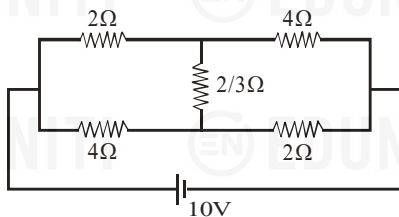


3. Relation between current in conductor and time is shown in figure.

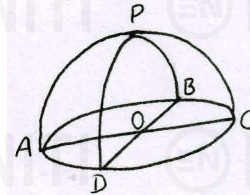


- (a) Find the total charge flow through the conductor
(b) Write the expression of current in terms of time
(c) If the resistance of conductor is R, then the total heat dissipated across resistance R is

4. Find the current through $(2/3)\Omega$ resistor in the figure shown.



5. A hemisphere network of radius 'a' is made by using a conducting wire of resistance per unit length r. Find the equivalent resistance across OP.

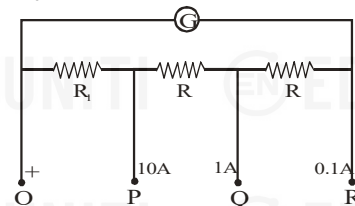


6. A total charge Q flows across a resistor R during a time interval T in such a way that the current v/s time graph for 0 to T is like the loop of a sine curve in the range 0 to π . What will be the total heat generated in the resistor.

7. A rod of length L and cross-section area A lies along the x-axis between $x = 0$ and $x = L$. The material obeys ohm's law and its resistivity varies along the rod according to $\rho(x) = \rho_0 e^{-x/L}$. The potential is V_0 at $x = 0$ and is zero at $x = L$.

- (a) Find the total resistance of the rod and the current in the rod.
(b) Find the electric potential in the rod as a function of x.

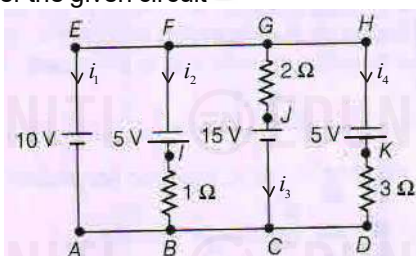
8. The resistance of the galvanometer G in the circuit is 25Ω . The meter deflects full scale for a current of 10 mA. The meter behaves as an ammeter of three different ranges. The range is 0–10A, if the terminals O and P are taken, the range is 0–1 A between O and Q and the range is 0–0.1 A between O and R. Calculate the resistances R_1 , R_2 and R_3 .



9. A galvanometer having 50 divisions and full scale deflection current 300 mA is provided with a variable shunt. It is used to measure the current as an ammeter when connected in series with a resistance of 90Ω and a battery of internal resistance 10Ω . It is observed that when the shunt resistances are 10Ω and 50Ω respectively, the deflections are respectively 9 and 30 divisions. What is the resistance of the galvanometer and the emf of the cell?

Passage (Q 10 to 12)

Consider the given circuit



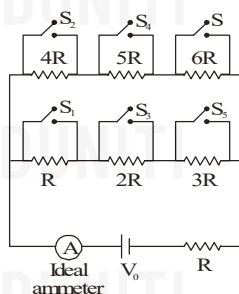
It is known that potential at point A is zero. Now, answer the following questions.

10. In the given circuit,
 - (a) one battery acts like source
 - (b) two batteries acts like source
 - (c) three batteries acts like source
 - (d) all batteries act like source
11. Which resistor consumes maximum power?
 - (a) 1 Ω resistor
 - (b) 2 Ω resistor
 - (c) 3 Ω resistor
 - (d) none of the above
12. Which battery consumes or gives maximum power?
 - (a) 10 V battery
 - (b) 15 V battery
 - (c) 5 V battery between points F and B
 - (d) 5 V battery between points H and D

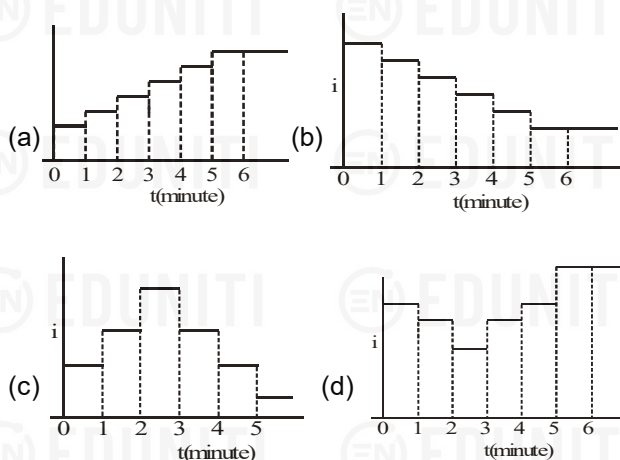
Passage 13 to 15

For the circuit shown, answer the following questions

13. In which of the following case, the current shown by ammeter is maximum
 - (a) S_1, S_2, S_3 closed
 - (b) S_2, S_4, S_5 closed
 - (c) S_1, S_3, S_5 closed
 - (d) S_2, S_3, S_4 closed



14. If switches S_1, S_2 and so on upto S_6 are closed at regular intervals of 1 minute starting from $t = 0$, the graph of current versus time is best represented as

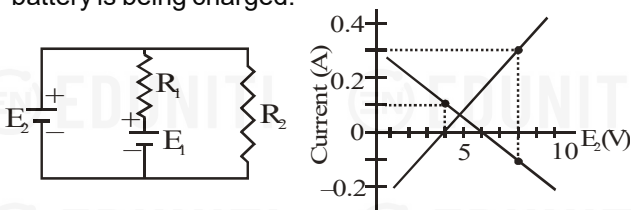


15. Ratio of power developed by battery when all switches are closed to that when all switches are open

- (a) $\frac{37}{7}$
- (b) $\frac{7}{37}$
- (c) $\left(\frac{37}{7}\right)^2$
- (d) $\left(\frac{7}{37}\right)^2$

Passage 16 to 18

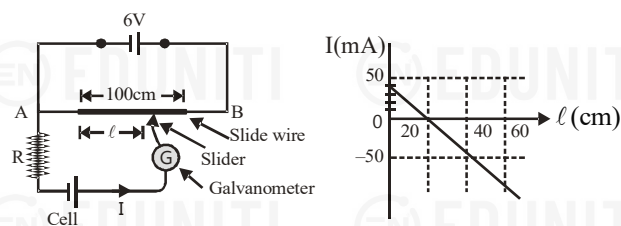
In the circuit shown, both batteries are ideal. Emf E_1 of battery 1 has a fixed value, but emf E_2 of battery 2 can be varied between 1 V and 10V. The graph gives the currents through the two batteries as a function of E_2 , but are not marked as to which plot corresponds to which battery. But for both plots, current is assumed to be negative when the battery is being charged.



16. The value of emf E_1 is
 - (a) 8V
 - (b) 6V
 - (c) 4V
 - (d) 2V
17. The resistance R_1 has value
 - (a) 10 Ω
 - (b) 20 Ω
 - (c) 30 Ω
 - (d) 40 Ω
18. The resistance R_2 is equal to
 - (a) 10 Ω
 - (b) 20 Ω
 - (c) 30 Ω
 - (d) 40 Ω

Passage 19 to 22

In the circuit shown, the internal resistance of the cell is negligible. The distance of the jockey (slider) from left hand end of the wire is l . The adjoining graph shows the variation of current I (marked in figure) with length l of the slide wire.



19. For balance condition of the instrument, the value of l is equal to
 - (a) 40 cm
 - (b) 20 cm
 - (c) 100 cm
 - (d) None of these
20. The emf of cell is
 - (a) 0.98 V
 - (b) 1.20 V
 - (c) 1.86 V
 - (d) 3 V

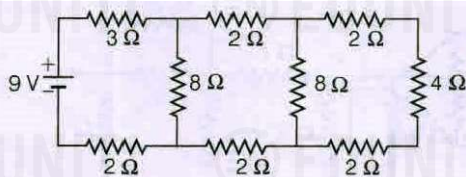
21. Value of the resistance R is

- (a) $30\ \Omega$ (b) $40\ \Omega$
(c) $38\ \Omega$ (d) $45\ \Omega$

22. The magnitude of current I when $l = 100\text{ cm}$ is

- (a) 50 mA (b) 120 mA
(c) 160 mA (d) 300 mA

23. In the circuit shown in the figure, the current through

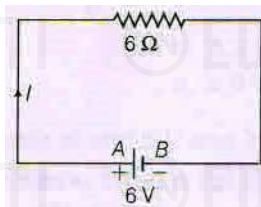


- (a) $3\ \Omega$ resistor is 0.5 A (b) $3\ \Omega$ resistor is 1 A
(c) $4\ \Omega$ resistor is 0.5 A (d) $4\ \Omega$ resistor is 0.25 A

24. 24 cells, each of emf 1.5 V and internal resistance $2\ \Omega$ are connected in mixed grouping so as to send the maximum current through a $12\ \Omega$ resistor. Then,

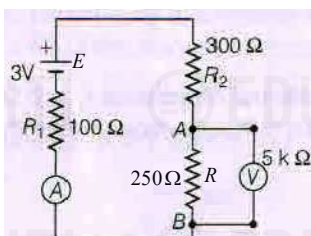
- (a) there are two rows of 12 cells in each row
(b) the current in each row is 0.375 A
(c) the current in each row is 0.75 A
(d) the potential difference across $12\ \Omega$ resistor is 9 V

25. In the given circuit, AB is a wire of length 10 cm and resistance $6\ \Omega$. The number of electrons per unit volume in material of wire is 10^{29} m^{-3} . If cross-sectional area of wire is 1 mm^2 , then which of the following options are correct



- (a) Energy absorbed by electrons is $0.9 \times 10^{-17}\text{ J}$
(b) Energy absorbed by electrons is $1.8 \times 10^{-17}\text{ J}$
(c) Ohmic loss in wire is 6 J/s
(d) All kinetic energy of electrons will be lost in around $3.0 \times 10^{-18}\text{ s}$

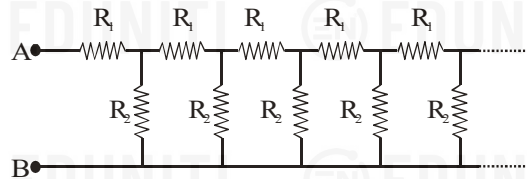
26. In given circuit, $E_1 = 3\text{ V}$, $R_1 = 100\ \Omega$, $R_2 = 300\ \Omega$ and $R_3 = 250\ \Omega$. Assume cell and ammeter as ideal. The voltmeter resistance is $5\text{ k}\Omega$.



- (a) Voltmeter reads 3% lower value of potential drop across R_3

- (b) Voltmeter reads 3% higher value of potential drop across R_3
(c) If V_{AB} is balanced against a potentiometer with potential gradient of 0.01 V/cm , then balance length is 115 cm
(d) If V_{AB} is balanced against a potentiometer with potential gradient of 0.01 V/cm , then balance length is 112 cm

27. Consider an infinite ladder network shown in figure. A voltage V is applied between the points A and B . This applied value of voltage becomes successively half after each section. Then

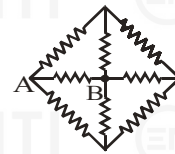


- (a) $\frac{R_1}{R_2} = 1$ (b) $\frac{R_1}{R_2} = \frac{1}{2}$
(c) $\frac{R_1}{R_2} = 2$ (d) $\frac{R_1}{R_2} = 3$

28. In a potentiometer wire experiment, the emf of a battery in the primary circuit is 20 V and its internal resistance is $5\ \Omega$. There is a resistance box in series with the battery whose resistance can be varied from $120\ \Omega$ to $170\ \Omega$. Resistance of the potentiometer wire is $75\ \Omega$. Which of the following potential differences can be measured using this potentiometer?

- (a) 5 V (b) 6 V
(c) 7 V (d) 8 V

29. In the circuit shown, each resistor has resistance $5\ \Omega$. The points A and B are connected to the terminals of a cell of emf 9 volt and internal resistance $(2/3)\ \Omega$.



- (a) The heat produced in the cell is 6 W .
(b) The current in the resistor connected directly between A and B is 1.4 A .
(c) The current in the resistor connected directly between A and B is 1.8 A .
(d) None of the above is correct.

30. In the circuit shown in figure,

- (a) power supplied by the battery is 200 watt
(b) current flowing in the circuit is 5 A
(c) potential difference across $4\ \Omega$ resistance is equal to the potential difference across $6\ \Omega$ resistance
(d) current in wire AB is zero

